Xinmei Yang

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Education

Harvard University — **Cambridge, MA, USA**
Ph.D. in Economics; Ph.D. minor in Data Science (GPA: 3.88/4.00) — *May 2028 (expected)*

- **Selected honors & awards:** Harvard Graduate Fellowship; Garon Economics Research Award ($10 000); Reviewer for Journal of Public Economics, SoFiE Financial Machine Learning Summer School (Yale SOM), Machine Learning in Economics Summer Institute (Chicago Booth), Incentives for Collaborative Learning and Data Sharing Workshop, Scheduled Presenter, 2026 INFORMS Annual Meeting.

Renmin University of China — **Beijing, China**
B.Sc. in Resource Economics (GPA: 3.97/4.00) — *Jun 2023*

- **Selected honors & awards:** National Academic Scholarship; CUMCM National First Prize; ICM Finalist; Fung Scholarship.

Programming Skills

Python, SQL, R, Stata, PyTorch, Detectron2, Transformers, Apache Airflow, Powerhouse (Roblox internal platform)

Relevant Professional Experience

Roblox Corporation — **San Mateo, CA**
Data Science Intern — *Jun 2026 – Aug 2026*

- Build daily Apache Airflow pipelines on Roblox's internal Powerhouse platform to integrate large-scale item, user and creator features, and recommendation-funnel datasets; develop Apache Superset dashboards to track cold-start item performance, decompose funnel-stage conversion, and identify retrieval as a key bottleneck limiting new item exposure.
- Develop a tree-based model using item embeddings and user, creator, and item-level features to shortlist high-potential cold-start items; apply visual-similarity search to filter copycat and near-duplicate marketplace listings.
- Evaluate model performance using t-tests and LLM-based judges, and collaborate with machine learning engineers and cross-functional partners to translate findings into A/B-test treatment logic, success metrics, and production implementation requirements for a planned online experiment.

Research Experience

Harvard University — **Cambridge, MA**
Doctoral Researcher and Coauthor — *Sep 2021 – present*

- Constructed a firm-level dataset from two historical Japanese books' Optical Character Recognition (OCR) results to investigate the institutional historical roots for economic growth in East Asian countries through network analysis.
- Ran Deep learning Mask R-CNN model using Detectron2 and MMDetection and fine-tuned the hyperparameters to predict layout and OCRed texts for books; created training data from label studio by converting between coco and label studio format dataset; prepared annotation materials for dedicated RAs to label.
- Developed a customized Python script to postprocess OCR results for a horizontal layout book; designed unit tests to verify validity of the output dataset.
- Worked on an example of homoglyph edit distance fuzzy matching using dynamic programming; designed a customized labeling interface in Jupyter notebook to conduct ground truth construction; fine-tuned custom hyperparameters for homoglyph edit distance.
- Developing a language-model training pipeline using text digitized from one publication year with vision-based layout detection to process historical publications with different layouts but similar language across years; building a temporal knowledge graph of firms, banks, and government officials and applying network motif analysis to study persistent relationships over time.

Publications & Preprints

1. **Xinmei Yang**, Michael Li & Robert Huckman. **The Value of Primary Care Beyond Prevention: Evidence from Intensive Care Unit Mortality**, 2026.
2. **Xinmei Yang***, Abhishek Arora, Shao-Yu Jheng & Melissa Dell. **Quantifying Character Similarity with Vision Transformers**, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
3. Abhishek Arora, **Xinmei Yang***, Shao-Yu Jheng & Melissa Dell. **Linking Representations with Multimodal Contrastive Learning**, 2023.
4. **Xinmei Yang***, Junru Lyn. **Does My Contribution Matter? Pivotality and Investment in Crowdfunding**, 2025.